26/11/19

Indian Institute of Engineering Science and Technology, Shibpur

B.Tech (CST) 3^{rd} Semester Final Examination, November 2019

Digital Logic (CS 301)

Answer any 5 questions.

F.M. 70 Time: 3 hrs

1. (a) Compare 1s and 2s complement number representation.

(b) Express the following Boolean function in a product of maxterm form.

$$F = xy + \overline{x}z$$

(c) Simplify the following Boolean function

$$F(w,\ x,\ y,\ z) = \Sigma(4,\ 5,\ 7,\ 12,\ 14,\ 15) + d(3, 8, 10).$$

(d) Define fan-out and noise-margin of a logic gate. [3 + 3 + 4 + 4]

2. (a) Which logic level limits the fan-out of TTL gates? Explain your answer with a suitable diagram.

(b) Draw the circuit diagram of a two inputs DTL NAND gate. Calculate fan-out and noise-margins of the DTL gate. The diode and transistor parameters are:
Diode: Voltage across a conducting diode = 0.7 V
Cut-in voltage = 0.6 V
Transistor: Cut-in voltage = 0.5 V, $V_{BE,sat} = 0.8V$
$V_{CE,sat} = 0.2V$
$h_{FE} = 30$ [4 + 10]

3. (a) Obtain the logic diagram that converts a four-digit binary number to a decimal number in BCD. Note that two decimal digits are needed since the binary numbers range from 0 to 15.

(b) Implement the following function with a multiplexer

$$F(A,\ B,\ C,\ D) = \Sigma(0,\ 1,\ 3,\ 4,\ 8,\ 9,\ 15)$$

[10 + 4]

4. (a) Construct a 5×32 decoder with four 3×8 decoder/demultiplexers and a 2×4 decoder. Use block diagram construction.

(b) A combinational circuit is defined by the following three functions:

$$F_1 = \overline{x}.\overline{y} + x.y.\overline{z}$$

$$F_2 = \overline{x} + y$$

$$F_3 = x.y + \overline{x}.\overline{y}$$

Design the circuit with a decoder and external gates. [6 + 8]

5. (a) Design a BCD ripple counter using JK flip-flops and also draw the timing diagram of the said counter.

(b) Design a counter with the following binary sequence:
1, 2, 5, 3, 7, 6, 4
and repeat. Use D flip-flops in the design process. [8 + 6]

6. (a) A full-adder receives two external inputs X and Y; the carry input Z comes from the output of a D flip-flop. The carry output is transferred to the flip-flop in every clock pulse. The external output S gives the sum of X, Y and Z. Obtain the state table and state diagram of the sequential circuit.

(b) Define state equation. When is the state equation method convenient for designing sequential circuits? What is the main difference between a sequential circuit and a combinational circuit? [7 + (2+ 3+ 20)]

7. (a) Why we do not use S = 1 and R = 1 as inputs to R-S (using NOR gates) flip-flop? What is the problem in J-K flip-flop and how is it removed?

(b) Draw the logic diagram of the positive edge-triggered D flip-flop and explain its operation. [4 + 10]

8. Define lockout of counters. Design a mod-5 counter using JK flip-flop so that if the unused states 101, 110 and 111 occur, the next clock pulse will reset the counter to 000. [3 + 11]